DAQQA (■■■■■■) • Spaceflight Health Monitoring

Telemetry Data Sources & Physiological Threshold Specification

Document ID: DAQQA-SPEC-HRP09-2026	**Mission Event:** 2026 NASA Space Apps Challenge
Software Track: Healthcare Track (Space Apps)	**Target Hardware:** Garmin tactix 7 / fenix 7 (Connect IQ 7)
Clinical Standards: NASA LSDA & LSAH Dual Protocol	**Temporal Sync:** Sub-second Co-Registration (Δt = 0 ms)

1. Executive Technical Summary

This specification establishes the definitive data provenance, sensory instrumentation channels, and mathematical decision thresholds implemented in the **DAQQA (■■■■■■)** astronaut health telemetry engine. Designed for long-duration deep space exploration (Artemis Lunar Gateway, Mars transit), DAQQA addresses communication blackouts (LOS) through autonomous on-wrist processing and provides flight surgeons with rigorous, standardized physiological risk classification.

2. Sensory Data Provenance & Acquisition Architecture

All telemetry streams within DAQQA originate from verified physical and behavioral interfaces running on Garmin's microgravity-tested wearable architecture:

Telemetry Channel	Hardware / Sensor Modality	Acquisition API / Runtime	Sampling & Provenance
Heart Rate (HR)	Garmin Elevate Gen 5 Optical PPG (Green 525nm)	`Toybox.Sensor.Info.heartRate`	Continuous 1 Hz photoplethysmogram tracking pulsatile blood volume.
Pulse Oximetry (SpO2)	Dual-Wavelength Optical LEDs (Red 660nm / IR 940nm)	`Toybox.Sensor.Info.oxygenSaturation`	Synchronously co-registered with HR at identical UTC epoch (Δt = 0 ms).
Heart Rate Variability	High-Frequency R-R Interbeat Intervals	`Toybox.SensorHistory (RMSSD)`	Quantifies vagal parasympathetic vs sympathetic autonomic tone.
Physical Workload (METs)	3-Axis Accelerometer (100 Hz microgravity inertial unit)	`Toybox.Sensor.registerSensorData`	Vector jerk integration yielding Metabolic Equivalents of Task (1.0 - 9.5).
Skin Temperature	Integrated Thermistor Lens Sensor	`Toybox.SensorHistory.getTemperature`	Monitors peripheral vasoconstriction, circadian rhythm, and thermal strain.
Behavioral Surveys	On-Wrist Interactive Task UI (5-Button Tactical Interface)	`Custom Monkey C View / Storage Module`	NASA-TLX, POMS mood/focus, and Karolinska Sleep Scale analog.
Offline NVRAM Buffer	Non-Volatile RAM (Garmin tactix 7 onboard storage)	`Toybox.Application.Storage`	Autonomous data queue holding up to 30 days during comm blackouts (LOS).

3. Physiological Thresholds & Clinical Decision Criteria

DAQQA employs multi-tiered, scientifically grounded decision boundaries across each physiological dimension:

Table 3.1: Pulse Oximetry (SpO2) & Cardio-Oxygen Saturation Index (COSI)

Parameter	Threshold Boundary	Classification	Operational & Clinical Meaning
SpO2 Saturation	≥ 95.0%	`NOMINAL`	Normal alveolar-capillary oxygen diffusion in cabin atmosphere.
SpO2 Saturation	90.0% – 94.9%	`MILD DESATURATION`	Elevated cabin altitude, hypoventilation, or early suit scrub deficiency.
SpO2 Saturation	< 90.0%	`CRITICAL HYPOXIA`	**Immediate Emergency Alarm:** Audible & haptic warning to check O2 umbilical.
COSI Ratio (SpO2 / HR × 100)	≥ 1.40	`OPTIMAL COUPLING`	Healthy cardiorespiratory transport efficiency.
COSI Ratio	< 1.00	`ACUTE MISMATCH`	Compensatory tachycardia in presence of falling oxygen saturation.

Table 3.2: Personalized Baseline Deviation Engine (z-Scores)

Standardized scale-invariant deviation formula: $z = (x_t - \mu_{baseline}) / \sigma_{baseline}$

Deviation Tier	z-Score Range	Clinical Assessment	Flight Surgeon Directive
Nominal Stability	`|z| ≤ 1.5 σ`	Physiological homeostatic equilibrium.	Nominal flight schedule.
Mild Drift	`1.5 < |z| ≤ 2.5 σ`	Sub-clinical physiological drift.	Monitor trend; check hydration and sleep telemetry.
Moderate Deviation	`2.5 < |z| ≤ 3.5 σ`	Statistically significant physiological strain.	Flag for ground review; recommend scheduled rest break.
Critical Anomaly	`|z| > 3.5 σ (or HR > 3.0σ)`	Severe acute pathological deviation.	Immediate medical triage; halt physical EVA workload.

Astronaut ID & Name	Resting HR (μ ± σ)	SpO2 Baseline (μ ± σ)	HRV RMSSD (μ ± σ)	Stress Baseline
ASTRO-01 (Dr. Elena Rostova - CDR)	62.4 ± 4.8 BPM	98.4 ± 0.6 %	68.2 ± 7.1 ms	22.0 ± 5.5 pts
ASTRO-02 (Marcus Vance - FE)	70.2 ± 5.4 BPM	97.6 ± 0.7 %	55.0 ± 6.8 ms	32.0 ± 6.2 pts
ASTRO-03 (Dr. Kenji Sato - MS)	58.1 ± 4.2 BPM	99.0 ± 0.5 %	74.0 ± 8.0 ms	18.5 ± 4.8 pts
ASTRO-04 (Amina Al-Mansoor - SP)	65.8 ± 4.6 BPM	98.1 ± 0.6 %	62.5 ± 6.5 ms	26.0 ± 5.2 pts

Table 3.3: Resting Heart Rate (RHR) & Cardiovascular Deconditioning

RHR elevation delta: $\Delta RHR = HR_{measured_nadir} - RHR_{baseline}$ (tracked during METs ≤ 1.8):

Resting HR Delta (ΔRHR)	Cardiovascular Status	Pathophysiological Mechanism in Microgravity
`ΔRHR ≤ +4 BPM`	**Stable Autonomic Baseline**	Normal baroreceptor function and fluid volume adaptation.
`+5 to +12 BPM`	**Mild Deconditioning / Latent Fatigue**	Plasma volume contraction or high sympathetic tone from sleep loss.
`≥ +15 BPM`	**Critical Deconditioning / Autonomic Strain**	Significant orthostatic intolerance risk; severe central volume loss.
`HR > 145 BPM (Absolute)`	**Sustained Tachycardia Alarm**	Exceeds safe operational sustained aerobic threshold; mandatory pause.

Table 3.4: HR vs. Activity Concordance & Cardiorespiratory Coupling

Karvonen Expected Heart Rate: *Expected_HR = RHR + (185 - RHR) × ((METs - 1.0) / 7.5)*
Exertion Strain Ratio (ESR): *ESR = (Actual_HR - RHR) / max(8, Expected_HR - RHR)*

Coupling Classification	Mathematical Trigger Criteria	Clinical Spaceflight Diagnosis
CONCORDANT	`|Actual_HR − Expected_HR| ≤ 12 BPM (ESR ≈ 1.0 ± 0.25)`	Optimal cardiorespiratory coupling. Heart rate scales proportionally with physical workload.
INAPPROPRIATE TACHYCARDIA (UNCOUPLED)	`METs ≤ 2.0 (Rest/Sedentary) AND Actual_HR > (RHR + 35 BPM)`	**CRITICAL EMERGENCY:** Acute hypoxia, airlock depressurization, panic, or cardiac arrhythmia occurring while astronaut is resting.
EXERTIONAL STRAIN DRIFT	`(Actual_HR − Expected_HR) > 22 BPM (ESR > 1.8)`	Excessive cardiac cost for light/moderate work. Indicates deconditioning, heat accumulation, or dehydration.
CHRONOTROPIC INCOMPETENCE	`METs ≥ 5.5 (Vigorous EVA Work) AND Actual_HR < (RHR + 20 BPM)`	Blunted chronotropic response. Cardiac output insufficient for high metabolic EVA demand.

Table 3.5: Sensor Reliability & Signal Quality Index (SQI)

SQI Range	Reliability Status	Gating Behavior & False Alarm Prevention
`≥ 85 %`	**HIGH_CONFIDENCE**	Optimal optical signal-to-noise ratio. Full diagnostic validity.
`65 % – 84 %`	**MODERATE_CONFIDENCE**	Microgravity motion detected. Band-pass filtering active; alarms trusted with weighting.
`45 % – 64 %`	**SUSPECT_ARTIFACT**	**Alarm Gating Active:** Suppresses false positive biometric alarms during intense workouts.
`< 45 %`	**SENSOR_DEGRADED**	Loose watch strap or detachment. Triggers on-wrist hardware fit warning.
`PI < 1.0 %`	**POOR PERFUSION**	Insufficient optical pulse amplitude (vasoconstriction or cold exposure).

Table 3.6: Multimodal Risk Score (MMRS) Composite Matrix

MMRS Score	Clinical Risk Category	Countermeasure Protocol
`0 – 29`	**Nominal Adaptation (Green)**	Routine spaceflight schedule and standard exercise countermeasure.
`30 – 59`	**Moderate Autonomic Strain (Amber)**	Recommend 30-minute relaxation pod session; check sleep architecture.
`60 – 79`	**High Autonomic Exhaustion (Orange)**	Electrolyte rehydration bolus; postpone planned spacewalk (EVA).
`80 – 100`	**Critical Decompensation (Red)**	Immediate medical officer intervention; high-flow O2 and vital resuscitation.

4. Scientific CSV Data Extraction Schema

The exported dataset complies strictly with **RFC 4180** and is encoded in **UTF-8 with Byte Order Mark (BOM)**. Every row contains synchronized readings at the identical timestamp epoch ($\Delta t = 0$ ms):

Category	CSV Column Fields Included
Temporal & Identity	`Synchronous_Timestamp_ISO_UTC, Unix_Epoch_Milliseconds, Astronaut_ID, Astronaut_Name, Flight_Day`
Cardiorespiratory	`Heart_Rate_BPM, SpO2_Percent, Cardio_Oxygen_Saturation_Index_COSI, HRV_RMSSD_ms, Respiration_Rate_BrPM`
Activity & Exertion	`Activity_Level_Category, Activity_METs, Activity_Intensity_Pct, Accelerometer_Magnitude_G`
Resting HR & Concordance	`Resting_HR_Baseline_BPM, Current_Resting_HR_BPM, HR_Elevation_Over_Resting_BPM, Expected_HR_For_Activity_BPM, Exertion_Strain_Ratio_ESR, Cardiorespiratory_Concordance_Status`
Reliability & SQI	`Signal_Quality_Index_SQI_Pct, Sensor_Reliability_Status, Perfusion_Index_Pct, Motion_Artifact_Level`
Clinical AI Risk & Prediction	`Multimodal_Risk_Score_MMRS, Clinical_Syndrome_Diagnosis, Predicted_Recovery_Index_24h, Deterioration_Probability_24h_Pct, Telemetry_Alarm_Status`
Behavioral Surveys	`Subjective_Sleep_Quality_Score, Subjective_POMS_Focus_Score`